

Software Categories

Lesson 18 — Types and Licensing Models

JKSSB Computer Awareness

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What Is Software?

- ▶ **Software** is a set of programs (instructions) that tell the computer what to do.
- ▶ Hardware is the **physical** part; software is the **logical** part of a computer system.
- ▶ Programs are stored in memory and executed by the **CPU**.
- ▶ Without software, hardware cannot perform any useful task.

The one idea

Hardware is what you can touch; software is the set of instructions that makes that hardware work.

The Main Software Categories

Category	One-line role
System software	Controls hardware and provides a platform for other software
Application software	Helps the user perform specific tasks
Utility software	Performs maintenance and housekeeping tasks
Programming software	Used to write and translate programs
Embedded software	Built into a device to control that device
Middleware	Connects different applications and systems

These categories overlap in use, but each answers a different question about “what the program is for”.

System Software

- ▶ **System software** manages and controls the computer hardware.
- ▶ It provides the **platform** on which application software runs.
- ▶ Examples: the operating system, device drivers, firmware (BIOS), and utility programs.
- ▶ It runs close to the hardware and usually starts when the machine is switched on.

Trap alert

A device driver is **software**, not hardware, and the operating system is **system software** (TRAP-10).

The Operating System

- ▶ The **operating system (OS)** is the main system software.
- ▶ It acts as an **interface** between the user and the hardware.
- ▶ It manages memory, files, processes, and input/output devices.
- ▶ Examples: Windows, Linux, macOS, and Android.

Why it matters

The OS is the first layer of software a program depends on; without it, an application cannot reach the hardware.

Application Software

- ▶ **Application software** helps the user perform a **specific task**.
- ▶ It runs on top of the system software.
- ▶ Examples: MS Word, MS Excel, web browsers, media players, and accounting packages.
- ▶ It is written for the user's work, not to manage the machine.

General-Purpose vs Specific-Purpose

General-purpose

- ▶ Ready-made packages sold to many users
- ▶ Examples: MS Word, MS Excel, a web browser
- ▶ Also called *application packages*

Specific-purpose (custom)

- ▶ Built for one organisation or one task
- ▶ Examples: a payroll system, a railway reservation system
- ▶ Also called *tailor-made* software

Utility Software

- ▶ **Utility software** is a small program that performs **maintenance** tasks.
- ▶ Examples: antivirus programs, disk cleanup, disk defragmentation, backup, and file compression.
- ▶ It keeps the system healthy, fast, and safe.
- ▶ It belongs to the system software family.

Trap alert

A utility **maintains the system**; an application helps the user do **work**. Do not call antivirus an application for producing documents.

Programming Software and Translators

- ▶ **Programming software** is used to write, translate, and test programs.
- ▶ **Language translators**: compiler, interpreter, and assembler.
- ▶ Supporting tools: linker, loader, debugger, and text editor.
- ▶ An **IDE** (Integrated Development Environment) combines an editor, a compiler, and a debugger in one tool.

Compilers and interpreters

A compiler translates the whole program at once; an interpreter translates and runs it line by line.

Embedded Software

- ▶ **Embedded software** is built into a device to control that device.
- ▶ It is usually stored in **ROM** as firmware.
- ▶ It runs automatically when the device is switched on.
- ▶ Examples: a washing machine, a microwave oven, a router, and a car engine-control unit.

Key point

Embedded software is hidden inside the device and is written for one dedicated job.

Middleware

- ▶ **Middleware** connects different applications and systems so they can work together.
- ▶ It sits between the operating system and application software.
- ▶ It provides common services such as messaging, authentication, and data exchange.
- ▶ Example: an application server that links a web front end to a database.

Licensing Models: Overview

Model	Key point
Proprietary	Source code closed; use governed by a licence
Open-source	Source code available to view, modify, and redistribute
Free software	Freedom to run, study, modify, and share
Freeware	Free of cost, but source may be closed
Shareware	Trial version; payment needed for full use
Public domain	No copyright; free for any use

Commercial software is sold for profit and is usually proprietary.

Proprietary vs Open-Source

Proprietary

- ▶ Source code is not published
- ▶ Licence restricts copying and modification
- ▶ Examples: MS Windows, MS Office

Open-source

- ▶ Source code is published
- ▶ Licence allows viewing, changing, and sharing
- ▶ Examples: Linux, Mozilla Firefox, LibreOffice

Note

“Open-source” refers to **source-code availability**, not to price.

Freeware, Shareware, Free Software, Public Domain

- ▶ **Freeware**: distributed free of cost; the source code may not be available.
- ▶ **Shareware**: offered as a limited trial; payment is needed for continued or full use.
- ▶ **Free software**: gives the user freedom to run, study, modify, and redistribute it (“free” as in freedom, not price).
- ▶ **Public-domain software**: has no copyright and may be used freely.

Trap Alert: Three Terms, Three Meanings

TRAP-23

Open-source $\neq$ freeware $\neq$ shareware.

- ▶ **Open-source**: the **source code** is available.
- ▶ **Freeware**: free of **cost**, but source may be closed.
- ▶ **Shareware**: a **trial**; pay to continue.

The nearest-neighbour confusion here is the single most common licensing distractor.

Lesson 18: Recall

- ▶ Software is a set of programs; hardware is the physical part.
- ▶ System software manages hardware (OS, device drivers, firmware); application software helps the user do work.
- ▶ Utility software performs maintenance; programming software writes and translates programs; embedded software controls a device; middleware connects systems.
- ▶ Licensing: proprietary (closed), open-source (source available), free software (four freedoms), freeware (free of cost), shareware (trial then pay), public domain (no copyright).
- ▶ TRAP-23: open-source, freeware, and shareware are **not** the same.